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Chapter 1

Introduction

You should be able to define and understand the following after finishing these notes.

- Complexity theory, universal computation models, universal turing machine, Church-Turing thesis and its other forms, hypercomputation, random algorithm and probabilistic Turing machine, probabilistic Church-Turing thesis, Deutsch notion of universal quantum computiong, discrete logarithm and Shor's algorithm, Grover's algorithm and space searching, error correcting codes, CSS codes, superdense coding, reversing zero channel, RSA and cryptosystems, Diaffie-Hallamn.
- Bell-EPR pair and no classical explanation, Maxwell's deamon, Szilards' engine adn exorcism.
- Reversible computation, billiard ball model, Fredkin, Conservative logic

Before any formal approach we give a brief introduction to quantum computing concepts and application. We model a *quantum state space* with a vector space (Hilbert space to be exact)[1]. Each vector represent a *quantum state* and can be thought of a *superposition* of some basis state. A quantum bit or *qubit* is an example of 2-dimensional quantum system with $|0\rangle$ and $|1\rangle$ as basis and a state $|\psi\rangle$ can be represented by

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad \alpha, \beta \in \mathbb{C}$$

We may also consider a normalization condition for such state. That is, we must have $|\alpha|^2 + |\beta|^2 = 1$. We can interpret this coefficients to be the probability of result of measuring $|\psi\rangle$. That is, $\mathbb{P}(M|\psi\rangle = |0\rangle) = |\alpha|^2$ and $\mathbb{P}(M|psi\rangle = |1\rangle) = |\beta|^2$. Because of the normalization condition, we can write $|\psi\rangle$ as

$$|\psi\rangle = e^{i\gamma}\left(\cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}e^{i\phi}|1\rangle\right)$$

dropping the $e^{i\gamma}$ factor allows us to represent the state space on a sphere called *Bloch sphere*.

Suppose we have two qubits then the basis of the system is $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ and every state is represent in a superposition of this basis

$$|\psi\rangle = \sum_{x\in\{0,1\}^2} \alpha_x|x\rangle \quad \text{with} \quad \sum_{x\in\{0,1\}^2} |\alpha_x|^2 = 1$$

[1] Non-linear behavior can lead to time-travel, faster than light communication, and violation of second law of thermodynamics

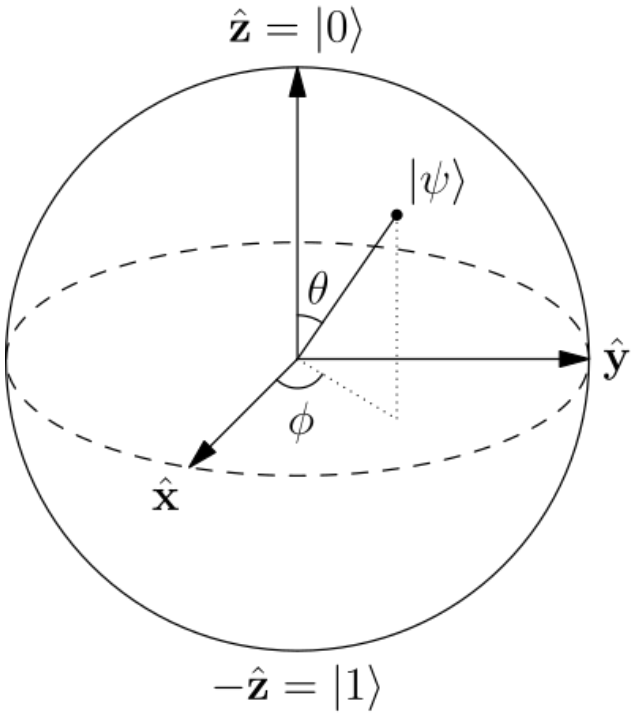

Figure 1.1: Bloch sphere

If we measure the first we get $|0\rangle$ with probability

$$|\alpha_{00}|^2 + |\alpha_{01}|^2$$

and the post-measurement state would be

$$|\psi'\rangle = \frac{\alpha_{00}|00\rangle + \alpha_{01}|01\rangle}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}}$$

Bell state or EPR pair is

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

A state of multiple qubit is called *separable* if each of the qubit is in a definite state i.e. we can write it as a tensor product. Otherwise, they are in an *entangled* state.

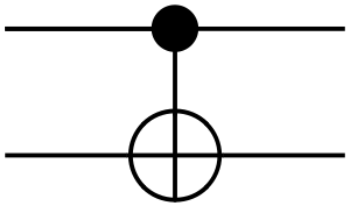

Figure 1.2: CNOT gate

1.1 Quantum gates and measurements

Isolated quantum mechanic processes (evolution) are represent by unitary matrices $UU^\dagger = I$. Some example of quantum gates include

$$\sigma_x = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{Quantum NOT gate}$$

$$\sigma_y = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad \text{Y gate}$$

$$\sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \text{Z gate}$$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \text{Hadamard gate}$$

The first three gate correspond to 180° rotation along the x, y, and z respectively, and are called the *Pauli rotation matrices*. Every gate on a qubit can be viewed as a set of rotations along the x, y, z-axis of the Bloch sphere. Furthermore, every unitary matrix can be viewed geomtrically as scaling-rotation-scaling. A *Controlled NOT* gate is a two qubit gate that given $|\psi\rangle$ and $|\phi\rangle$ input, gives $|\psi\rangle$ and $|\psi \oplus \phi\rangle$. The matrix of CNOT gate is

$$U_{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Quantum gates need to be reversible, that is given the output one can find out the output. For example, classical NOT gate is reversible and XOR gate is non-reversible. All multiple qubit gate may be decomposed to CNOT and other single qubit gates. Feedback, FANIN (irreversible) , FANOUT (cloning) are not allowed in quantum computing. In general, Controlled-U gate is shown as and has matrix representation

$$\text{Controlled}U = \begin{bmatrix} I_n & 0 \\ 0 & U \end{bmatrix}$$

and measurement of a quantum state is represent by

No cloning theorem states that lossless copying of qubit using unitary devices can obly be done on orthogonal basis. That is, if $|\psi\rangle$ can be copied to $|\phi\rangle$ then

$$\langle\phi|\psi\rangle = \begin{cases} 0 & \psi \perp \phi \\ 1 & \psi = \phi \end{cases}$$

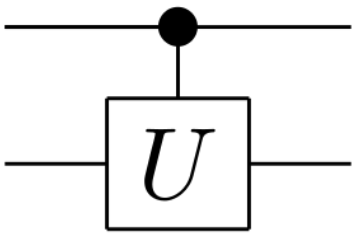

Figure 1.3: Controlled-U gate

Figure 1.4: Measurement

1.2 Quantum teleportation

The other Bell states are

$$
\begin{aligned}
|\beta_{00}\rangle &= \frac{|00\rangle + |11\rangle}{\sqrt{2}} \\
|\beta_{01}\rangle &= \frac{|01\rangle + |10\rangle}{\sqrt{2}} \\
|\beta_{10}\rangle &= \frac{|00\rangle - |11\rangle}{\sqrt{2}} \\
|\beta_{11}\rangle &= \frac{|01\rangle - |10\rangle}{\sqrt{2}}
\end{aligned}
$$

which are generated by the following circuit –insert diagram Hadamard on the frist qubit and CNOT on the second with matrix

$$
U = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix} = \begin{bmatrix} |\beta_{00}\rangle & |\beta_{01}\rangle & |\beta_{10}\rangle & |\beta_{11}\rangle \end{bmatrix}
$$

Suppose Alice and Bob have an EPR pair $|\beta_{00}\rangle$, and each took one the pairs. Alice then wants to communicate to Bob the state $|\psi\rangle$ using only classical bit. Alice and Bob can implement the following – insert diagram CNOT on Alice qubit, Hadamard on psi, measure psi and alice

qubit, not gate on bob and measured alice and Z gate on bob and measure psi.

$$
\begin{aligned}
|\psi_0\rangle &= \frac{\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle)}{\sqrt{2}} \\
|\psi_1\rangle &= \frac{\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|10\rangle + |01\rangle)}{\sqrt{2}} \\
|\psi_2\rangle &= \frac{\alpha|+\rangle(|00\rangle + |11\rangle) + \beta|-\rangle(|00\rangle + |11\rangle)}{\sqrt{2}} \\
&= \frac{1}{2}|00\rangle(\alpha|0\rangle + \beta|1\rangle) + \frac{1}{2}|01\rangle(\beta|0\rangle + \alpha|1\rangle) \\
&+ \frac{1}{2}|10\rangle(\alpha|0\rangle - \beta|1\rangle) + \frac{1}{2}|11\rangle(\beta|0\rangle - \alpha|1\rangle) \\
&= \frac{1}{2}|00\rangle|\psi\rangle + \frac{1}{2}|01\rangle|X\psi\rangle + \frac{1}{2}|10\rangle|Z\psi\rangle + \frac{1}{2}|11\rangle|XZ\psi\rangle
\end{aligned}
$$

Therefore, after measurements and applying X and Z gates, Bob will have $|\psi\rangle$. Quantum teleportation is related to quantum erroc correcting codes.

1.3 Reversible computing

Can Quantum computer simulat classical circuits? Yes; since any classical circuit can be replaced by reversible elements such as Toffoli gate (–insert diagram for Toffoli gate, NAND , and FANOUT). The matrix representation of Toffoli gate

$$
U = \begin{bmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
1 & 0 & 0 & 0 & 0 & 0 & 1 & 0
\end{bmatrix} = \begin{bmatrix} I_6 & 0 \\ 0 & X \end{bmatrix}
$$

Note that FANOUT creates entangled copies of the quantum state hence it does not violate the no cloning theorems. We can even assume that classical computer can make random bit. It is easy to see that by using Hadamard gate and then measuring it, quantum computers can make random bits. –insert diagram

1.4 Quantum Algorithms and parallelism

Let $f : \{0,1\} \to \{0,1\}$ and U_f be $|x, y\rangle \mapsto |x, f(x) \oplus y\rangle$.

$$
U_f = \begin{bmatrix}
f'(0) & f(0) & 0 & 0 \\
f(0) & f'(0) & 0 & 0 \\
0 & 0 & f'(1) & f(1) \\
0 & 0f(1) & f'(1) &
\end{bmatrix}
$$

–insert diagram of Uf and Hadamard Applying $|+,0\rangle$ to U_f gives

$$\begin{aligned}
U_f|+,0\rangle &= \frac{1}{\sqrt{2}}U_f|00\rangle + \frac{1}{\sqrt{2}}U_f|10\rangle \\
&= \frac{1}{\sqrt{2}}(f'(0)|00\rangle + f(0)|01\rangle) + \frac{1}{\sqrt{2}}(f'(1)|10\rangle + f(1)|11\rangle) \\
&= \frac{|0,f(0)\rangle + |1,f(1)\rangle}{\sqrt{2}}
\end{aligned}$$

Hence, with a single operation we can "evaluate" f for all values. Using Hadamard-Welsh Transform we can do this for any function.

1.4.1 Deutsch Algorithm

Let $f : \{0,1\}^n \to \{0,1\}$ with U_f. Similar to parallelism is one-bit case we can have

$$U_f|0\ldots0,0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle \otimes |f(x)\rangle$$

using inference property/ measure (will be like the classical case) we can extract information about f. For example, in Deutsch algorithm we use U_f and construct the following –insert diagram

$$\begin{aligned}
|\psi_0\rangle &= |01\rangle \\
|\psi_1\rangle &= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \\
|\psi_2\rangle &= U_f|\psi_1\rangle \\
&= \frac{1}{2}\begin{bmatrix} f'(0) - f(0) \\ f(0) - f'(0) \\ f'(1) - f(1) \\ f(1) - f'(1) \end{bmatrix}
\end{aligned}$$

let $p_0 = f'(0) - f(0)$ and $p_1 f'(1) - f(1)$ then

$$\begin{aligned}
&= \frac{1}{2}(p_0|00\rangle - p_0|01\rangle + p_1|10\rangle - p_1|11\rangle) \\
&= \frac{1}{2}(p_0|0\rangle + p_1|1\rangle) \otimes (|0\rangle - |1\rangle) \\
&= \begin{cases} (-1)^{f(0)}|+\rangle \otimes |-\rangle & f(0) = f(1) \\ (-1)^{f(0)}|-\rangle \otimes |-\rangle & f(0) \neq f(1) \end{cases} \\
|\psi_3\rangle &= \begin{cases} (-1)^{f(0)}|0\rangle \otimes |-\rangle & f(0) = f(1) \\ (-1)^{f(0)}|1\rangle \otimes |-\rangle & f(0) \neq f(1) \end{cases} \\
&= (-1)^{f(0)}|f(0) \oplus f(1)\rangle \otimes |-\rangle
\end{aligned}$$

hence in one operation we can learn a global property of the function namely the value of $f(0) \oplus f(1)$. A generalization of Deutsch algorithm is the Deutsch-Jozsa algorithm which

extends $f : \{0,1\}^n \to \{0,1\}$. First, consider the n-qubit Hadamard gate is defined as

$$\begin{aligned} H^{\otimes n}|x_1, \ldots, x_n\rangle &= H|x_1\rangle \otimes \cdots \otimes H|x_n\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{x \cdot z} |z\rangle \end{aligned}$$

where $x \cdot z = x_1 z_1 + x_2 z_2 + \cdots + x_n z_n$. In particular

$$\begin{aligned} H^{\otimes n}|0 \ldots 0\rangle &= \frac{1}{\sqrt{2^n}} (|0\rangle + |1\rangle) \otimes \cdots \otimes (|0\rangle + |1\rangle) \\ &= \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \end{aligned}$$

The Deutsch-Jozsa algorithm works as follow –insert diagram

$$\begin{aligned} |\psi_0\rangle &= |0 \ldots 0, 1\rangle \\ |\psi_1\rangle &= |+ \cdots +, -\rangle = \frac{1}{\sqrt{2^n}} \left(\sum_{x \in \{0,1\}^n} |x\rangle \right) \otimes |-\rangle \\ |\psi_2\rangle &= \frac{1}{\sqrt{2^n}} \left(\sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle \right) \otimes |-\rangle \\ |\psi_3\rangle &= \frac{1}{2^n} \left(\sum_{z \in \{0,1\}^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x) + x \cdot z} |z\rangle \right) \otimes |-\rangle \end{aligned}$$

Now we can determine whether f is constant or balance (has the same number of one and zeros) by measuring the first qubit.

- If f is constant then the amplitude of the first bit is 1.
- If f is balanced then the amplitude of the first bit is 0.

Hence after measuring if $|0\rangle$ then f is constant, otherwise f is balanced.

Chapter 2

Linear Algebra

Vectors in quantum mechanic are denoted by $|\psi\rangle$. Its conjugate transpose is $\langle\psi| = |\psi\rangle^\dagger$ and we can define a inner product $\langle\phi,\psi\rangle = \langle\phi|\psi\rangle$. Given an inner product we can define norm

$$\||\psi\rangle\| = \sqrt{\langle\psi|\psi\rangle}$$

The Pauli matrices are $\mathcal{M}_2(\mathbb{C})$

$$\sigma_0 = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \sigma_1 \qquad\qquad = \sigma_x = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\sigma_2 = \sigma_y = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \qquad \sigma_3 = \sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

For $|v\rangle \in V, |w\rangle \in W$ we can define an outer product $|w\rangle\langle v| : V \to W$.

$$|w\rangle\langle v||v'\rangle = \langle v|v'\rangle|w\rangle$$

The completeness relation say that given an orthogonal basis $|i\rangle$

$$\sum |i\rangle\langle i| = I$$

which is easy to see

$$\sum |i\rangle\langle i||v\rangle = \sum \langle i|v\rangle|i\rangle = v$$

Given an inner product vector space we can define the Cauchy-Schwarz inequality

$$\langle v|v\rangle\langle w|w\rangle \geq |\langle v|w\rangle|^2$$

2.1 Adjoint and Hermitian operators

For any linear operator on a Hilbert space there exists B such that for all $v, w \in V$

$$\langle v|Aw\rangle = \langle Bv|w\rangle$$

It is easy to see that $B = A^\dagger$ the adjoint or Hermitian conjuage of A. $A = A^\dagger$ is a Hermitian or self-adjoint operator. A projection is an operator that projects $|v\rangle$ into its compenents on a subspace. Suppose $|1\rangle, \ldots, |k\rangle$ is an orthonormal basis for subspace W

$$P = \sum_{i=1}^{k} |i\rangle\langle i|$$

Geoemtrically, applying P twice to a vector should again give the projection of that vector that is, $P^2 = P$.

$$\begin{aligned}
P^2 &= \left(\sum_{i=1}^{k} |i\rangle\langle i|\right)\left(\sum_{j=1}^{k} |j\rangle\langle j|\right) \\
&= \sum_{i=1}^{k}\sum_{j=1}^{k} |i\rangle\langle i||j\rangle\langle j| \\
&= \sum_{i=1}^{k}\sum_{j=1}^{k} |i\rangle\langle i|j\rangle\langle j| \\
&= \sum_{i=1}^{k} |i\rangle\langle i|i\rangle\langle i| \\
&= \sum_{i=1}^{k} |i\rangle\langle j| = P
\end{aligned}$$

$AA^\dagger = A^\dagger A$ is a normal operator. An operator is diagonalizable if it has a diagonal representation

$$A = \sum \lambda_i |i\rangle\langle i|$$

where λ_i are the eigenvalues and $|i\rangle$ form an orthonormal set for the eigenvectors of A. If an eigenspace has dimension greater than one then those eigenvectors are called degenerates.

Theorem 2.1 (Spectral decomposition). *Any normal operator M is diagonal with respect to orthonormal basis in V. The converse is also true, any diagonalizable matrix is normal.*

Proof. Lets induct over $\dim V$. If $\dim V = 1$ then it is trivial that M is diagonal. Suppose λ is an eigenvalue of M and P is the projection onto its eigenspace. Then, $M = (P+Q)M(P+Q) = PMP + PMQ + QMP + QMQ$ where $Q = I - P$. Clearly, $PMP = \lambda P$ and $QMP = 0$. We claim that $PMQ = QM^\dagger P$ is zero as well. Suppose $|v\rangle \in P$ then

$$M\big(M^\dagger|v\rangle\big) = M^\dagger M|v\rangle = \lambda M^\dagger|v\rangle$$

Therefore, $M^\dagger \in P$ as well, hence $PMQ = 0$. We then show that QMQ is normal as well.

$$\begin{aligned}
(QMQ)(QMQ)^\dagger &= QMQQM^\dagger Q \\
&= QMQM^\dagger Q \\
&= QMM^\dagger Q \qquad (QM^\dagger = QM^\dagger Q + QM^\dagger P = QM^\dagger Q) \\
&= QM^\dagger MQ \\
&= QM^\dagger QMQ \qquad (QMQ = MQ - PMQ = MQ) \\
&= (QMQ)^\dagger (QMQ)
\end{aligned}$$

Now note that PMP is diagonal with respect to an orthonormal basis for P and by induction hypothesis there is a basis for Q such that QMP is diagonal. Together, these two imply that M with respect to the union of these two basis is diagonal in V. Furthermore, this implies that M can be written as

$$M = \sum \lambda_i |i\rangle\langle i|$$

where λ_i are its eigenvalues. To show that converse, suppose M is diagonalizable. Then,

$$\begin{aligned} M^\dagger M &= \left(\sum \lambda_i^* |i\rangle\langle i|\right)\left(\sum \lambda_i |i\rangle\langle i|\right) \\ &= \sum_i \sum_j \lambda_i^* |i\rangle\langle i|\lambda_j|j\rangle\langle j| \\ &= \sum_i \|\lambda\|^2 |i\rangle\langle i| \end{aligned}$$

and similarly

$$\begin{aligned} MM^\dagger &= \left(\sum \lambda_i |i\rangle\langle i|\right)\left(\sum \lambda_i^* |i\rangle\langle i|\right) \\ &= \sum_i \sum_j \lambda_i |i\rangle\langle i|\lambda_j^*|j\rangle\langle j| \\ &= \sum_i \|\lambda\|^2 |i\rangle\langle i| \implies MM^\dagger = M^\dagger M \end{aligned}$$

Theorem 2.2. *A normal operator is hermitian if and only if it has real eigenvalues.*

Proof. A hermitian operator is normal and has real eigenvalues since

$$\langle v|A|v\rangle = \lambda\langle v|v\rangle$$

and

$$\langle v|A|v\rangle = \langle v|A^\dagger|v\rangle = (\langle v|A|v\rangle)^\dagger \implies \lambda = \lambda^*$$

Suppose A is a normal with real eigenvalues. By spectral decomposition

$$A^\dagger = \left(\sum \lambda_i |i\rangle\langle i|\right)^\dagger = \sum \lambda_i^* |i\rangle\langle i| = \sum \lambda_i |i\rangle\langle i| = A$$

$UU^\dagger = I$ is unitary.

Proposition 2.3. *an unitary operator*

1. *preserves inner product.*
2. *there are two orthonormal basis $|v_i\rangle, |w_i\rangle$ such that*

$$U = \sum |w_i\rangle\langle v_i|$$

3. *its eigenvalues have modulus* 1

Proof. 1.

$$\langle Uv|Uw\rangle = \langle v|U^\dagger U|w\rangle = \langle v|w\rangle$$

2. Let $|v_i\rangle$ be an orthonormal set and $|w_i\rangle = U|v_i\rangle$. Then, by above's result $|w_i\rangle$ are orthonormal as well. Also note that for any $v \in V$, $\langle v_i|v\rangle = (w_i)Uv$. Then,

$$\sum |w_i\rangle\langle v_i||v\rangle = \sum |w_i\rangle\langle v_i|v\rangle = \sum |w_i\rangle\langle w_i|Uv\rangle = \sum |w_i\rangle\langle w_i||Uv\rangle = U|v\rangle$$

therefore

$$U = \sum |w_i\rangle\langle v_i|$$

3. Let $(v, |v\rangle)$ be a pair of eigenvalue and eigenvector of U.

$$\begin{aligned}\langle Uv|Uv\rangle &= \langle v|v\rangle \\ &= \|v\|^2\langle v|v\rangle \implies \|v\| = 1\end{aligned}$$

A is positive when for all $|v\rangle$, $\langle v|A|v\rangle \geq 0$ and positive definite if the equality only happens when $|v\rangle = 0$.

Proposition 2.4. *Any operator A can be written as $A = B + iC$ where B, C are hermitian.*

Proof. Let

$$B = \frac{A + A^\dagger}{2} \qquad C = \frac{A - A^\dagger}{2i}$$

then clearly $A = B + iC$ and both of the hermitian.

$$B^\dagger = \frac{A^\dagger + A}{2} = B \qquad C^\dagger = \frac{A^\dagger - A}{-2i} = C$$

Proposition 2.5. *Positive operators are hermitian.*

Proof. By the last result $A = B + iC$ where B, C are hermitian. Since B, C are hermitian then they have spectral decomposition

$$B = \sum \lambda_i |v_i\rangle\langle v_i| \qquad C = \sum \gamma_j |w_j\rangle\langle w_j|$$

where λ_i, γ_j are real numbers. For every $|v\rangle \in V$

$$\begin{aligned}\langle v|B|v\rangle &= \sum \lambda_i \langle v||v_i\rangle\langle v_i||v\rangle \\ &= \sum \lambda_i \|\langle v_i|v\rangle\|^2\end{aligned}$$

is a real number. Therefore, since

$$\langle v|A|v\rangle = \langle v|B|v\rangle + i\langle v|C|v\rangle$$

is a real number as well then

$$\langle v|C|v\rangle = 0 \qquad \forall |v\rangle$$

Then for any $(\gamma_j, |w_j\rangle)$

$$\langle w_j|C|w_j\rangle = \gamma_j\langle w_j|w_j\rangle = 0$$

since w_j are orthonormal then $\gamma_j = 0$ and hence $C = 0$. ■

2.2 Tensor product

Let $|v\rangle \in V$ and $|w\rangle \in W$ then

$$|v\rangle \otimes |w\rangle = \begin{bmatrix} v_1w_1 & v_1w_2 & \dots & v_1w_m \\ \vdots & \vdots & \ddots & \vdots \\ v_nw_1 & v_nw_2 & \dots & v_nw_m \end{bmatrix} \qquad V \otimes W = \{|v\rangle \otimes |w\rangle \,|\, |v\rangle \in V, |w\rangle \in W\}$$

If $|i\rangle$ and $|j\rangle$ are orthonormal basis for V and W then $|i\rangle \otimes |j\rangle$ is a basis for $V \otimes W$. For operators

$$(A \otimes B)(|v\rangle \otimes |w\rangle) = A|v\rangle \otimes B|w\rangle$$

with the Kroneker matrix representation

$$A \otimes B = \begin{bmatrix} A_{11}B & A_{12}B & \dots & A_{1n}B \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1}B & A_{m2}B & \dots & A_{mn}B \end{bmatrix}$$

If A is $m \times n$ and B is $p \times q$ then $A \times B$ is $mp \times nq$.

Proposition 2.6. *1. Tensor product of unitary operators is unitary.*

2. Tensor product of hermitian operators is hermitian.

3. Tensor product of projection operators is projection.

4. Tensor product of positive operator is positive.

2.3 Operator function

Let A be a normal operator then

$$f(A) = \sum f(\lambda_i)|i\rangle\langle i|$$

trace of a matrix is the sum of its diagonal elements.

$$\operatorname{tr} A = \sum a_{ii}$$

Proposition 2.7. *1. it is commutative*

$$\operatorname{tr} AB = \operatorname{tr} BA$$

2. it is linear

$$\operatorname{tr} A + cB = \operatorname{tr} A + c \operatorname{tr} B$$

3. it is invariant under unitary transformation

$$\operatorname{tr} UAU^\dagger = \operatorname{tr} A$$

Proposition 2.8. *Let $\mathcal{L}(V)$ be all the linear function on V. Give a basis for $\mathcal{L}(V)$ and show that*

$$\langle A, B\rangle = \operatorname{tr} A^\dagger B$$

is an inner product. Find a basis for hermitian matrices for $\mathcal{L}(V)$.

2.4 Commutators and anti-commutators

$$[A, B] = AB - BA \qquad \{A, B\} = AB + BA$$

Theorem 2.9 (Simultaneous diagonalization theorem). *Suppose A and B are hermitian. They commute if and only if there exists an orthonormal basis such that A and B are both diagonalizable with respect to that basis.*

2.5 Polar decomposition and SVD

Theorem 2.10. *Let A be a linear operator on V. There exists an unitary operator and positive operators J and K such that*

$$A = UJ = KU$$

UJ is the left PD and KU is the right PD. J, K are unique and defined by $J = \sqrt{A^\dagger A}$ and $L = \sqrt{AA^\dagger}$. If A is invertible then U is unique.

Theorem 2.11. *Let A be a square matrix. There are unitary matrices U, V and diagonal matrix D with non-negative entries such that*

$$A = UDV$$

Proof. By PD, $A = SJ$ and from spectral theorem $J = TDT^\dagger$ where T is unitary and D is diagonal. Therefore $A = UDT^\dagger$ where $U = ST$. ■

Chapter 3

Quantum Mechanic

3.1 Axioms of quantum mechanic

Each physical system is a seperable complex Hilbert – complete vector space – with inner product $\langle\psi, \phi\rangle$. Rays – complex subspaces of dimesion 1 – in $\mathcal{H}$ are associated with quantum state of the system. We bring an incomplete set of quantum mechanic axioms.

Definition:

Postulate I: The state of an isolated physical system at a fixed time t is represented by a (unit) stated vector $|\psi\rangle$ belongin to $\mathcal{H}$.

Postulate II: The evolution of a closed quantum system is described by a unitary transformation.

$$|\psi_{t_1}\rangle = U|\psi_{t_0}\rangle$$

The time evolution of the state of a closed quantum system is described by Schrodinger equation.

$$ih\frac{\mathrm{d}}{\mathrm{d}t}|\psi\rangle = H|\psi\rangle$$

Where H is the Hamiltonian operator. Hamiltonian is a hermitian operator – $H = H^\dagger$– and it can be decomposed to its energy levels (eigenvalues).

$$H = \sum_E E|E\rangle\langle E|$$

Postulate III: Quantum measurements are described by a collection $\{\mathcal{M}_m\}$ of measurement operators satisfying the completeness relation

$$\sum \mathcal{M}_m^\dagger \mathcal{M}_m = I$$

These are operators acting on the state space of the system being measured.

$$\mathbb{P}(m) = \langle\psi|\mathcal{M}_m^\dagger \mathcal{M}_m|\psi\rangle$$

is the probability of measuring m. $\{\mathcal{M}_m\}$ are basically the eigenvectors of a hermitian operator – therefore, $\mathcal{M}_m^\dagger \mathcal{M}_m$ is the eigenspace. The stated post measurement is

$$\frac{\mathcal{M}_m|\psi\rangle}{\sqrt{\langle\psi|\mathcal{M}_m^\dagger \mathcal{M}_m|\psi\rangle}}$$

Postulate IV: The composite state space is the tensor product of the state spaces of the component physical systems.

Remark 1. Non-orthogonal states can not be reliably distinguished. Suppose there is a measurement device that can distinguish non-orthogonal states $|\psi_1\rangle, |\psi_2\rangle$. Suppose $|\psi_b\rangle$ is prepared, then the probability of measuring j such that $f(j) = b$ is 1. Define

$$E_i = \sum_{j;f(j)=i}$$

3.2 Projective measurements

A projective measurement is described by an observable, M, a hermitian operator on the state space of the system being observed.

$$M = \sum mP_m$$

where P_m is the projectors into eigenspace with $P_iP_j = \delta_{ij}P_i$. Then, the probability of getting result m is

$$\mathbb{P}(m) = \langle\psi|P_m|\psi\rangle$$

We define the average and variance of a projective measurement as follows.

$$\begin{aligned}
\langle M\rangle = \mathbb{E}[M] &= \sum m\mathbb{P}(m) \\
&= \sum m\langle\psi|P_m|\psi\rangle \\
&= \langle\psi|\Big(\sum mP_M\Big)|\psi\rangle \\
&= \langle\psi|M|\psi\rangle(\Delta M)^2 && = \big\langle(M - \langle M\rangle)^2\big\rangle \\
&= \big\langle M^2\big\rangle - \langle M\rangle^2
\end{aligned}$$

Remark 2 (Heisenberg uncertainty principle). Suppose $|\psi\rangle$ is a quantum state and A, B are hermitian operators. Let

$$x + iy = \langle\psi|AB|\psi\rangle$$

then,

$$\langle\psi|BA|\psi\rangle = \langle\psi|(AB)^\dagger|\psi\rangle = (\langle\psi|AB|\psi\rangle)^\dagger = x - iy$$

therefore,

$$|\langle\psi|[A, B]|\psi\rangle| = 2|x| \leq 2|\langle\psi|AB|\psi\rangle|$$

With Cauchy-Schwarz inequality

$$\begin{aligned}
|\langle\psi|AB|\psi\rangle|^2 &\leq \langle\psi|A^2|\psi\rangle\langle\psi|A^2|\psi\rangle \\
\implies |\langle\psi|[A, B]|\psi\rangle| &\leq 4\langle\psi|A^2|\psi\rangle\langle\psi|A^2|\psi\rangle
\end{aligned}$$

Hence if we let $A = C - \langle C\rangle, B = D - \langle D\rangle$ then

$$\begin{aligned}
[A, B] &= [C, D] \\
\big\langle A^2\big\rangle &= (\Delta C)^2, \big\langle B^2\big\rangle = (\Delta D)^2
\end{aligned}$$

and

$$(\Delta C)(\Delta D) \geq \frac{|\langle\psi|[C,D]|\psi\rangle|}{2}$$

Which basically means that if two measurements C, D do not commute then as the error in measuring one decreases the error in measuring the other one must increase. Hence, there would always be an uncertainty in the exact properties of the system.

Let $\vec{v}$ be a direction in $\mathbb{R}^3$ then, the measurement of spin along $\vec{v}$ is defined as

$$\vec{v} \cdot \vec{\sigma} = v_1\sigma_1 + v_2\sigma_2 + v_3\sigma_3$$

where σ_i are the Pauli matrices.

3.3 POVM measurement

Suppose $\mathcal{M}_m$ are measurement operators. Then,

$$E_m = \mathcal{M}_m^\dagger \mathcal{M}_m$$

are positive and complete – $\sum_m E_m = I$ –. The complete set of $\{E_m\}$ is called "Positive Operator Valued Measure" or POVM. We can get the $\{\mathcal{M}_m\}$ from $\{E_m\}$ by letting $\mathcal{M}_m = \sqrt{E_m}$.

Example 3.1. Suppose we want to distinguish between $|\psi\rangle = |0\rangle$ and $|\psi_2\rangle = |+\rangle$ with no error. Since these two states are

3.4 Density operator

Suppose a quantum system is prepared in one of the $|\psi_i\rangle$ states with probability p_i. The density operator for the system is

$$\rho = \sum p_i |\psi_i\rangle\langle\psi_i|$$

If the system evolves with unitary matrix U then the density operator evolves to

$$p = \sum p_i |\psi_i\rangle\langle\psi_i| \xrightarrow{U} \sum p_i U|\psi_i\rangle\langle\psi_i|U^\dagger = UpU^\dagger$$

Furthermore, if $\{\mathcal{M}_m\}$ are a set of measurements then,

$$\mathbb{P}(m \mid i) = \langle\psi_i|\mathcal{M}_m^\dagger \mathcal{M}_m|\psi_i\rangle = \operatorname{tr}\big(\mathcal{M}_m^\dagger \mathcal{M}_m|\psi_i\rangle\langle\psi_i|\big)$$

and

$$\begin{aligned}
\mathbb{P}(m) &= \sum p_i \mathbb{P}(m \mid i) \\
&= \sum p_i \operatorname{tr}\big(\mathcal{M}_m^\dagger \mathcal{M}_m|\psi_i\rangle\langle\psi_i|\big) \\
&= \operatorname{tr}\Big(\mathcal{M}_m^\dagger \mathcal{M}_m \sum p_i|\psi_i\rangle\langle\psi_i|\Big) \\
&= \operatorname{tr}\big(\mathcal{M}_m^\dagger \mathcal{M}_m \rho\big)
\end{aligned}$$

If m was measured in $|\psi\rangle$ then the post measurement state is

$$|\psi_i^m\rangle = \frac{\mathcal{M}_m|\psi\rangle}{\sqrt{\operatorname{tr}\Big(\mathcal{M}_m^\dagger\mathcal{M}_m|\psi_i\rangle\langle\psi_i|\Big)}}$$

and the density operator post measurement is

$$\begin{aligned}
\rho_m &= \sum \mathbb{P}(i \mid m)|\psi_i^m\rangle\langle\psi_i^m| \\
&= \sum \left(\frac{p_i \operatorname{tr}\big(\mathcal{M}_m^\dagger\mathcal{M}_m|\psi_i\rangle\langle\psi_i|\big)}{\operatorname{tr}\Big(\mathcal{M}_m^\dagger\mathcal{M}_m\rho\Big)}\right)\left(\frac{\mathcal{M}_m|\psi_i\rangle\langle\psi_i|\mathcal{M}_m^\dagger}{\operatorname{tr}\Big(\mathcal{M}_m^\dagger\mathcal{M}_m|\psi_i\rangle\langle\psi_i|\Big)}\right) \\
&= \frac{1}{\operatorname{tr}\Big(\mathcal{M}_m^\dagger\mathcal{M}_m\rho\Big)}\sum p_i\mathcal{M}_m|\psi_i\rangle\langle\psi_i|\mathcal{M}_m^\dagger \\
&= \frac{\mathcal{M}_m\rho\mathcal{M}_m^\dagger}{\operatorname{tr}\Big(\mathcal{M}_m\rho\mathcal{M}_m^\dagger\Big)}
\end{aligned}$$

Theorem 3.1. *An operator ρ is the density operator associated to some ensemble $\{p_i, |\psi_i\rangle\}$ if and only if it satisfies the following conditions*

1. $\operatorname{tr}(\rho) = 1$.
2. *ρ is a positive operator.*

We can reform the quantum mechanic postulate for density operator as follows.

Definition:

Postulate I: The state of an isolated physical system at a fixed time t is completely described by its density operator.

Postulate II: The evolution of a closed quantum system is described by a unitary transformation.

$$\rho_{t_1} = U\rho_{t_0}U^\dagger$$

Postulate III: Quantum measurements are described by a collection $\{\mathcal{M}_m\}$ of measurement operators satisfying the completeness relation

$$\sum \mathcal{M}_m^\dagger\mathcal{M}_m = I$$

These are operators acting on the density operator of the system being measured.

$$\mathbb{P}(m) = \operatorname{tr}\big(\mathcal{M}_m\rho\mathcal{M}_m^\dagger\big)$$

is the probability of measuring m. $\{\mathcal{M}_m\}$ are basically the eigenvectors of a hermitian operator – therefore, $\mathcal{M}_m^\dagger\mathcal{M}_m$ is the eigenspace. The stated post measurement is

$$\frac{\mathcal{M}_m\rho\mathcal{M}_m^\dagger}{\operatorname{tr}\Big(\mathcal{M}_m\rho\mathcal{M}_m^\dagger\Big)}$$

Postulate IV: The composite density operator is the tensor product of the density operator of the component physical systems.

$$\rho = \rho_1 \otimes \rho_2 \otimes \cdots \otimes \rho_n$$

The mean of an operator over a system described by ρ is

$$\begin{aligned}\langle A \rangle &= \sum p_i \langle \psi_i | A | \psi_i \rangle \\ &= \sum p_i \operatorname{tr}(A|\psi_i\rangle\langle\psi_i|) \\ &= \operatorname{tr}(A\rho)\end{aligned}$$